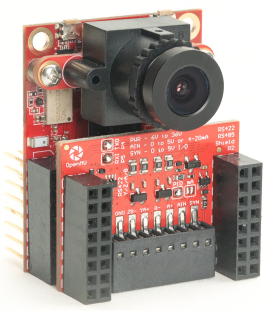
OpenMV® RS422/RS485 Shield

Datasheet

SKU: OMV-SHIELD-RS422-RS485

Description

The RS422/RS485 Shield allows your OpenMV Cam to transmit data to RS422 devices or communicate on an RS485 bus. YA+ and ZD- are the RS485 bidirectional pins (or RS422 receiver pins) while A+ and B- are the RS422 transmitter pins. The shield seamlessly handles either RS422 or RS485 (not both simultaneously). Transceivers connect to P4 (TX) and P5 (RX). Includes 6\u201336 V wide-range input, robust ADC, and robust trigger I/O.



Target Areas

Industrial communication, building automation, long-distance serial, multi-drop networks, PLC connectivity, SCADA systems.

Key Features

- **RS422 or RS485** — Seamlessly handles either protocol. RS422: separate TX (A+/B-) and RX (YA+/ZD-). RS485: bidirectional on YA+/ZD-.
- **6–36 V wide-range input** — Reverse voltage and transient surge protection. Outputs 5.4 V on VIN via OR'ing diode.
- **Robust ADC input** — 0–5 V on P6 with ±36 V overvoltage protection. Optional 4–20 mA mode.
- **Robust trigger I/O** — 0–5 V on P10 via SYN pin. Tolerant to -12 V, shorts to GND and 12 V.

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1 Features

1.1 Communication

Component	Details
RS422	Full-duplex RS422 with separate transmitter (A+/B-) and receiver (YA+/ZD-) pairs. Connected to P4 (TX) and P5 (RX). Disconnectable.
RS485	Half-duplex RS485 bidirectional on YA+/ZD- pins. Connected to P4 (TX) and P5 (RX). Cannot use RS422 and RS485 simultaneously.

1.2 Power Input

Component	Details
Wide-Range Input	6–36 V with reverse voltage and transient surge protection. Outputs 5.4 V on VIN via OR'ing diode.

1.3 ADC Input

Component	Details
AIN	Robust 0–5 V ADC input on P6 (0–3.3 V output). Overvoltage protection to ±36 V. Optional 4–20 mA mode. Disconnectable.

1.4 Digital I/O

Component	Details
SYN / P10	Robust 0–5 V digital I/O. Tolerant to -12 V, shorts to GND and 12 V. Default: open-drain input. Configurable as output. Disconnectable.

1.5 Compatibility

Component	Details
Supported Cameras	All OpenMV Cam models.

2 General Info

Dimension	Value
Weight	10 g

3 Company Information

Field	Details
Website	https://openmv.io

4 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/rs422-rs485-shield
Documentation	https://docs.openmv.io

5 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.